Tight Proximity Bounds for Multiple Group Labelings of SIBO Matroids

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Abstract

Let M be a matroid. For each $t \in [k]$, its elements are labeled by an abelian group Γ_t, and a finite set $F_t \subseteq \Gamma_t$ is forbidden. A basis is avoiding if its label sum is outside F_t for every labeling. Given any basis A, the *proximity problem for multiple group-label constraints* asks how many elements must be replaced to obtain an avoiding basis, assuming one exists.

Here this study determines the exact bound for subsequence-interchangeably base orderable (SIBO) matroids. If $m_t = |F_t|$, then there is an avoiding basis B with $|A \setminus B| \leq \prod_{t=1}^{k}(m_t+1)-1$. The bound is best possible even for uniform matroids with finite cyclic labeling groups. In particular, when each labeling forbids one value, the exact bound is $2^k - 1$. For this singleton setting, Garamvölgyi, Mizutani, Oki, Schwarcz, and Yamaguchi [5] proved a SIBO upper bound $\lfloor (e - \frac{1}{2})k! \rfloor - 1$ and a uniform-matroid lower bound $2^k - 1$. Our bound precisely matches their lower bound, closing the gap for SIBO matroids and answering the SIBO case of their Question 6.2.

The proof uses an exact bound for ordered sequences in products of abelian groups. Character interpolation and a triangular evaluation matrix give the bound, which we then apply to consecutive basis exchanges. The same argument gives proximity bounds for matroids over a fixed finite field and for sparse paving matroids. We further extended our existential result with a direct local-search algorithm. We also give a lower bound on the number of avoiding bases.

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1 Introduction

Matroid bases provide a common framework for combinatorial structures such as spanning trees and maximal linearly independent sets. In many problems, however, independence and cardinality are only part of the feasibility conditions. The selected elements must also satisfy arithmetic restrictions on their total labels. Integer labels can exclude specified totals, while labels in cyclic groups describe restrictions on residues. When several labelings are present, these requirements must be satisfied simultaneously.

Let M be a matroid on a finite ground set E. For each $t \in [k]$, let $\psi_t : E \to \Gamma_t$ be a labeling to an abelian group, and let $F_t \subseteq \Gamma_t$ be a finite set of forbidden values. For $X \subseteq E$, write $\psi_t(X) := \sum_{e \in X} \psi_t(e)$. A basis B is *avoiding* if $\psi_t(B) \notin F_t$ for every $t \in [k]$.

Thus feasibility requires avoidance in every coordinate. For example, a $\mathbb{Z}_2$-labeling with forbidden value 0 requires an odd total, while a second labeling may exclude several specified integer totals. The groups may differ between coordinates and need not be finite.

Our focus is the distance from an arbitrary basis to an avoiding one. Suppose that at least one avoiding basis exists. How many elements of a given basis A must be replaced to obtain an avoiding basis? For two bases A and B, we measure this by the *replacement distance* $d(A, B) := |A \setminus B| = |B \setminus A|$. The proximity question can be stated as follows.

Proximity Problem for Multiple Group-Label Constraints

Input: A finite matroid M, a basis A, and, for each $t \in [k]$, an abelian-group labeling $\psi_t : E(M) \to \Gamma_t$ and a finite forbidden set $F_t \subseteq \Gamma_t$.

Assumption: At least one avoiding basis exists.

Question: How small can a universal bound $g(m_1, \ldots, m_k)$ be, where $m_t = |F_t|$, such that every starting basis A has an avoiding basis B satisfying $d(A, B) \leq g(m_1, \ldots, m_k)$?

Such a bound controls the replacement distance using the arithmetic constraints rather than the size or rank of the matroid. It also has an algorithmic interpretation: an avoiding basis can be found by searching a bounded neighborhood of any starting basis.

Multiple labelings introduce a difficulty that is absent from the one-label setting. Although the maps ψ_t can be combined into a labeling to $\Gamma_1 \times \cdots \times \Gamma_k$, the resulting forbidden set is

$$\mathcal{F} := \bigcup_{t=1}^{k} \Gamma_1 \times \cdots \times \Gamma_{t-1} \times F_t \times \Gamma_{t+1} \times \cdots \times \Gamma_k.$$

This set can be infinite even when every F_t is finite. When all groups are finite, its cardinality can still depend on their orders. Consequently, treating the instance as a single labeling does not directly yield a proximity bound depending only on $|F_1|, \ldots, |F_k|$. The coordinate structure must be retained.

For singleton forbidden sets, Garamvölgyi et al. [5] gave a SIBO upper bound $\lfloor (e - \frac{1}{2})k! \rfloor - 1$ and a uniform-matroid lower bound $2^k - 1$, leaving a gap. Motivated by this gap, we close it completely for SIBO matroids and extend the exact bound to arbitrary finite forbidden sets.

We study this question primarily for *subsequence-interchangeably base orderable* (SIBO) matroids. This exchange property allows two bases to be ordered in paired positions so that exchanging any consecutive block produces another basis. It is weaker than strong base orderability, which permits exchanges of arbitrary subsets of paired elements. Consecutive exchanges are sufficient for our argument: their label changes are differences of prefix sums. This connection allows an algebraic bound on ordered sequences to determine the exact replacement radius.

1.1 Related Work

Group constraints on matroid bases have been studied both as *target-sum* requirements and as *avoidance* requirements. In the target-sum setting, Liu and Xu [8] study congruency-constrained bases, where the sum of integer labels must equal a specified residue modulo m. They develop closeness and strong closeness properties relating bases with prescribed label sums, and obtain fixed-parameter algorithms in the modulus for prime powers and products of two primes. Their framework connects additive structure in the labeling group with the exchange structure of a matroid. Our problem uses inequalities in each coordinate, and its proximity parameter is the number of labelings; the groups themselves need not be finite.

Proximity also supports algorithms for exact matroid problems with target-sum constraints. Eisenbrand, Rohwedder, and Węgrzycki [3] prove bounds for bases with prescribed sums of bounded integer vectors and derive fixed-parameter algorithms in the vector dimension and entry bound. They also obtain a fixed-parameter algorithm for a prescribed sum in a finite abelian group, parameterized by the group order. These results demonstrate how proximity can support algorithms whose polynomial exponent is independent of the parameter. Our structural bound depends only on k and applies to arbitrary abelian groups, but the direct neighborhood search derived here has an exponent depending on k. Accordingly, we state it as an XP algorithm and make no fixed-parameter tractability claim in the number of labelings.

Turning to avoidance constraints, Hörsch, Imolay, Mizutani, Oki, and Schwarcz [6] investigate group-labeled bases and common bases, including bases whose sum avoids a finite set F. They propose the proximity conjecture that, whenever an F-avoiding basis exists, every basis has an F-avoiding basis at replacement distance at most $|F|$. They establish the conjectured bound for $|F| \leq 2$, for ordered groups, and for strongly base orderable matroids. Their full version observes that the same argument applies to SIBO matroids [7, Remark 5.16]. These results concern a single labeling with several forbidden values. The present setting instead allows several different labelings, each with one forbidden value, and requires avoidance in every coordinate simultaneously.

These avoidance results make the exchange structure of the matroid central to the proximity question. The relevant ordering structure originates in Baumgart's work on ranking and ordering spanning trees [1]. Subsequence-interchangeable orderings make every consecutive exchange admissible. Strong base orderability imposes the stronger requirement that every subset of paired elements can be exchanged. Thus strongly base orderable matroids are SIBO, while the proof here uses only consecutive blocks. This distinction identifies the structural hypothesis needed by our argument: the algebraic reduction does not require arbitrary subsets of pairs to be exchangeable.

For multiple labelings within this exchange framework, the closest predecessor is Garamvölgyi, Mizutani, Oki, Schwarcz, and Yamaguchi [5]. They develop the proximity theory for group-labeled matroids and study simultaneous avoidance under k labelings. Their Theorem 5.5(1) gives the SIBO upper bound $\lfloor (e - \frac{1}{2})k! \rfloor - 1$, whereas their Example 5.1 gives a lower bound of $2^k - 1$ for uniform matroids. Their Question 6.2 asks whether $2^k - 1$ is always sufficient, explicitly highlighting uniform, strongly base orderable, and SIBO matroids. They also prove the bound 3 for two labelings in arbitrary matroids. Our result closes this gap for SIBO matroids and their subclasses. The uniform-matroid lower bound is an existing result; we give a self-contained version using finite cyclic groups.

1.2 Our Contributions and Overview

Here we give an overview of our proximity bounds and their consequences for finding and counting avoiding bases. Our main result determines the exact universal proximity radius for SIBO matroids with several finite forbidden sets. Writing $m_t := |F_t|,\ D := \prod_{t=1}^{k}(m_t+1)$, we show that $D-1$ replacements always suffice whenever an avoiding basis exists. A matching construction shows that no smaller universal radius is possible for these forbidden-set sizes.

▶ **Theorem 1** (Exact proximity). *Let M be a SIBO matroid, let $\psi_t : E(M) \to \Gamma_t$ be labelings to abelian groups, and let $F_t \subseteq \Gamma_t$ be finite for $t \in [k]$. Put $D := \prod_{t=1}^{k}(|F_t|+1)$. If an avoiding basis exists, then every basis A has an avoiding basis B with $|A \setminus B| \leq D-1$. For every choice of positive forbidden-set sizes, this bound is best possible, even for uniform matroids and labelings to finite cyclic groups.*

The theorem unifies the one-label and multiple-label settings. For one labeling, it gives $D-1 = |F_1|$, recovering the known one-label SIBO bound. When each labeling forbids one value, it gives $D-1 = 2^k-1$. Thus, for labelings ψ_t with forbidden values f_t, every starting basis A has a basis B satisfying $\psi_t(B) \neq f_t$ for every t and $|A \setminus B| \leq 2^k-1$, provided such a basis exists. This bound is best possible even for uniform matroids which is shown in Corollary 19.

This singleton result answers the SIBO case of Question 6.2 of [5], and therefore also its strongly base orderable and uniform cases. It replaces their factorial SIBO upper bound by 2^k-1, precisely matching their lower bound. Coordinates with empty forbidden sets impose no restriction and may be deleted; if all forbidden sets are empty, the radius is 0.

The proof begins with an exact algebraic statement about ordered sequences. In Theorem 12, we consider points $p_0, \ldots, p_{N-1}$ in a product of abelian groups such that, for every $a < b$, some coordinate satisfies $p_b^{(t)} - p_a^{(t)} \in C_t$, where C_t is a prescribed finite set with $0 \notin C_t$ and $|C_t| = m_t$. We prove that $N \leq \prod_{t=1}^{k}(m_t+1) = D$, and show that this bound is attainable.

The character-interpolation Lemma 11 supplies, in each coordinate, a function that equals 1 at 0, vanishes on the forbidden differences, and uses at most m_t+1 characters. Products of these functions lie in a space of dimension at most D. Their translates yield a triangular evaluation matrix with diagonal entries 1, forcing N linearly independent functions and hence $N \leq D$. Only finitely generated subgroups are needed, so the argument applies to arbitrary abelian labeling groups.

Prefix sums convert this sequence theorem into the exact interval-avoidance threshold of Corollary 15. Starting from an avoiding set, any collection of at least D ordered pairs of disjoint exchange blocks contains a nonempty consecutive exchange that preserves avoidance.

To obtain the SIBO upper bound, we choose an avoiding basis B closest to the given basis A. An SI-ordering of their differing elements makes every nonempty consecutive exchange a basis closer to A. If $|A \setminus B| \geq D$, the interval result produces an avoiding basis closer than B, contradicting its choice. This gives the replacement radius $D-1$.

The interval argument also applies to compatible exchanges of larger blocks. The Proposition 20 shows that weak (α, D)-base orderability implies a proximity radius of $\alpha-1$. Applying the known finite-field exchange theorem gives the radius $h(q,D)-1$ for $\mathrm{GF}(q)$-representable matroids, where h is the structural function (Corollary 21). For singleton forbidden sets, this becomes $h(q,2^k)-1$.

The corresponding exchange property for sparse paving matroids gives the following explicit consequence.

▶ **Corollary 2** (Sparse-paving proximity). *Let M be a sparse paving matroid with abelian-group labelings ψ_t and finite forbidden sets F_t, and put $D = \prod_{t=1}^{k}(|F_t|+1)$. If an avoiding basis exists, then every basis A has an avoiding basis B satisfying $|A \setminus B| \le 2^D - 1$. For k singleton forbidden sets, the bound is $2^{2^k} - 1$.*

These consequences extend the product threshold beyond the SIBO setting. The finite-field and sparse-paving bounds are not asserted to be optimal.

We complement the proximity results with a direct local-search algorithm. Suppose a matroid class has proximity radius d for the given constraints. For a rank-r matroid on n elements, put $R := \min\{d, r, n-r\}$, $S_R(r,n) := \sum_{i=0}^{R} \binom{r}{i}\binom{n-r}{i}$. The quantity $S_R(r,n)$ counts all r-element sets within distance R of a starting basis. Enumerating this neighborhood gives the following guarantee.

▶ **Lemma 3** (Bounded-radius search). *Suppose the input matroid belongs to a class with proximity radius d for the given constraints. Given a starting basis A, an independence oracle, group addition, inversion and equality, and explicit lists of the forbidden sets, there is an algorithm that finds an avoiding basis or certifies that none exists using at most $S_R(r,n)$ independence-oracle queries. Writing $L = \sum_{t=1}^{k}|F_t|$, the algorithm uses $O\big(kr + k(R+1)S_R(r,n)\big)$ group operations and at most $LS_R(r,n)$ equality tests against forbidden values.*

The algorithm enumerates nearby sets and tests their feasibility; it does not require computing characters or SI-orderings. For SIBO matroids, we take $d = D - 1$ (Corollary 24). Since $S_R(r,n) \le (R+1)(n+1)^{2R}$ and $D \le 2^L$, this search is XP in L in the stated oracle model. For singleton constraints, it is XP in k, with running time $(n+1)^{O(2^k)}$. The full search analysis is given in Proposition 23.

Finally, the same neighborhood bound gives a lower bound on the number of avoiding bases.

▶ **Corollary 4** (Counting avoiding bases). *Let M belong to a matroid class with proximity radius d for the given constraints. Let $\mathcal{B}(M)$ be its family of bases and let $\mathcal{B}_{\mathrm{av}} \neq \varnothing$ be its avoiding bases. With R and $S_R(r,n)$ defined above, $|\mathcal{B}_{\mathrm{av}}| \ge \frac{|\mathcal{B}(M)|}{S_R(r,n)}$. For SIBO matroids, one may take $R = \min\{D-1, r, n-r\}$.*

Thus the proximity theorem both locates an avoiding basis near every starting basis and bounds the proportion of bases that are avoiding. The counting argument is given in Corollary 25.

Organization. Section 2 gives the definitions and exchange properties. Section 3 proves the ordered-point and interval bounds. Section 4 proves the exact SIBO radius and gives worked examples. Section 5 gives the finite-field and sparse-paving consequences. Section 6 presents the search algorithm and counting bound. Section 7 concludes.

2 Preliminaries

For a nonnegative integer k, let $[k] = \{1, \dots, k\}$. All groups are abelian and written additively, with identity 0. We write $\mathbb{Z}_m = \mathbb{Z}/m\mathbb{Z}$ for the cyclic group of order m. For a labeling $\psi : E \to \Gamma$ and $X \subseteq E$, set $\psi(X) = \sum_{e \in X} \psi(e)$, $\psi(\varnothing) = 0$.

2.1 Matroids and Avoiding Bases

A finite matroid M consists of a ground set E and a nonempty family $\mathcal{B}(M)$ of bases satisfying the basis-exchange axiom: for bases A, B and $a \in A \setminus B$, there is $b \in B \setminus A$ such

that $(A \setminus \{a\}) \cup \{b\}$ is a basis. All bases have the same size, the *rank* r of M. A set is independent if it is contained in a basis. The uniform matroid $U_{r,n}$ has all r-element subsets of an n-element ground set as its bases. We use standard matroid terminology as in Oxley [9].

▶ **Definition 5** (Avoiding basis). *Let $\mathcal{F} = (F_1, \ldots, F_k)$, where each $F_t \subseteq \Gamma_t$ is finite, and let $\psi_t : E \to \Gamma_t$ be labelings. A basis B is* $\mathcal{F}$-avoiding*, or simply* avoiding*, if $\psi_t(B) \notin F_t$ for every $t \in [k]$. An instance is* feasible *if it has an avoiding basis.*

For bases A, B, their replacement distance is $\operatorname{dist}(A, B) = |A \setminus B| = |B \setminus A|$. Thus the symmetric difference has size $2\operatorname{dist}(A, B)$. For a feasible instance, the proximity question asks for a bound on the minimum distance from a given basis to an avoiding basis. An empty forbidden set imposes no restriction and may be deleted. If all forbidden sets are empty, every basis is avoiding and the radius is zero. We therefore assume $m_t = |F_t| \geq 1$ when discussing lower bounds.

For a class $\mathcal{C}$ of matroids, let $d_{\mathcal{C}}(m_1, \ldots, m_k)$ be the smallest nonnegative integer, if it exists, that bounds this distance for every feasible instance in $\mathcal{C}$ with $|F_t| = m_t$ and every starting basis. In the singleton case, abbreviate $d_{\mathcal{C}}(1, \ldots, 1)$ to $d_{\mathcal{C}}(k)$.

2.2 SIBO Orderings

The ordering property below follows the definition used by Baumgart [1] and Garamvölgyi et al. [5].

▶ **Definition 6** (SIBO). *A rank-r matroid is* subsequence-interchangeably base orderable *(SIBO) if, for every ordered pair of bases (A, B), there are orderings $a_1, \ldots, a_r$ of A and $b_1, \ldots, b_r$ of B such that $(B \setminus \{b_i, \ldots, b_j\}) \cup \{a_i, \ldots, a_j\}$ is a basis for all $1 \leq i \leq j \leq r$. Such a pair of orderings is called an* SI-ordering*.*

A matroid is *strongly base orderable* (SBO) if every pair of bases A, B admits a bijection $\phi : B \to A$ such that $(B \setminus X) \cup \phi(X)$ is a basis for every $X \subseteq B$. Ordering the elements of B and their images gives an SI-ordering. Hence every SBO matroid is SIBO.

We will exchange only elements on which two bases differ. The next lemma shows that common elements can be removed from the orderings.

▶ **Lemma 7** (Ordering the differing elements). *Let A, B be bases of a SIBO matroid and let $\ell = |A \setminus B|$. There are orderings $B \setminus A = \{x_1, \ldots, x_\ell\}$, $A \setminus B = \{y_1, \ldots, y_\ell\}$ such that $B_{i,j} = (B \setminus \{x_i, \ldots, x_j\}) \cup \{y_i, \ldots, y_j\}$ is a basis whenever $1 \leq i \leq j \leq \ell$.*

Proof. Choose an SI-ordering of (A, B). If $a_u \in A \cap B$ but $b_u \neq a_u$, then $(B \setminus \{b_u\}) \cup \{a_u\}$ has only $r - 1$ elements. This contradicts the singleton-interval condition. Thus every common element is paired with itself.

Delete the positions containing these common pairs and let $q_1 < \cdots < q_\ell$ be the remaining positions. Set $x_i = b_{q_i}$ and $y_i = a_{q_i}$. For $i \leq j$, apply the original SI-ordering to the interval $[q_i, q_j]$. Every deleted position in this interval exchanges an element with itself. The resulting set is therefore exactly $B_{i,j}$, so it is a basis. ◀

2.3 Weak Base Orderability and Other Matroid Classes

For later consequences, we need a weaker exchange condition in which whole blocks may be exchanged.

▶ **Definition 8** (Weak base orderability). *For positive integers α, s, a matroid is* weakly (α, s)-base orderable *if every ordered pair of bases (A, B) with $|A \setminus B| \geq \alpha$ has nonempty sets $X_1, \ldots, X_s \subseteq B \setminus A$, $Y_1, \ldots, Y_s \subseteq A \setminus B$, pairwise disjoint within each family, such that*

$$\left(B \setminus \bigcup_{i \in Z} X_i\right) \cup \bigcup_{i \in Z} Y_i$$

is a basis for every $Z \subseteq [s]$. It is elementarily *weakly (α, s)-base orderable if these sets can be chosen as singletons.*

Taking $Z = \{i\}$ shows that $|X_i| = |Y_i|$. This equality will ensure that every nonempty exchange moves closer to A.

A matroid is *representable over a field* if its elements correspond to columns of a matrix over that field and its independent sets correspond to linearly independent column sets. For a prime power q, let $\mathrm{GF}(q)$ denote the field with q elements. A rank-r matroid is *paving* if every circuit has size at least r, where a circuit is a minimal dependent set. It is *sparse paving* if both it and its dual are paving. The dual has bases $E \setminus B$ for $B \in \mathcal{B}(M)$; see also Bonin [2].

We use the following published exchange results.

▶ **Theorem 9** (Known weak exchange properties).

1. *There is a computable function $h : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that every $\mathrm{GF}(q)$-representable matroid is weakly $(h(q, s), s)$-base orderable for every positive integer s [6, 5].*
2. *Every sparse paving matroid is elementarily weakly $(\binom{2s}{s}, s)$-base orderable, and also elementarily weakly $(2^s, s)$-base orderable [5, Corollary 5.9 and Remark 5.10].*

The first statement is the finite-field theorem of Hörsch et al., recorded as Theorem 5.6 in [5]. We may take $h(q, s)$ nondecreasing in s by replacing it with $\max_{1 \leq u \leq s} h(q, u)$. These structural results are used only in Section 5.

3 Ordered Points and Avoiding Intervals

We first prove the group-theoretic statement used by all the proximity arguments. No matroid assumption is needed in this section.

3.1 Character Interpolation

A *character* of an abelian group H is a homomorphism $\chi : H \to \mathbb{C}^\times$, where $\mathbb{C}^\times$ is the multiplicative group of nonzero complex numbers. Thus $\chi(x + y) = \chi(x)\chi(y)$, where $\chi(0) = 1$, and $\chi(-x) = \chi(x)^{-1}$. We use this algebraic definition; character values need not have absolute value one. For a finite set S, the vector space $\mathbb{C}^S$ consists of all functions $S \to \mathbb{C}$, with pointwise addition and scalar multiplication. Its dimension is $|S|$.

The first lemma separates two distinct group elements.

▶ **Lemma 10** (Characters separate points). *If H is a finitely generated abelian group and $x, y \in H$ are distinct, there is a character $\chi : H \to \mathbb{C}^\times$ with $\chi(x) \neq \chi(y)$.*

Proof. Put $c = x - y \neq 0$. By the structure theorem for finitely generated abelian groups [4], $H \cong \mathbb{Z}^a \oplus \mathbb{Z}_{q_1} \oplus \cdots \oplus \mathbb{Z}_{q_b}$, $q_j \geq 2$. The element c has a nonzero component. If this component lies in a copy of $\mathbb{Z}$, compose the projection with $z \mapsto 2^z$. If it lies in $\mathbb{Z}_{q_j}$, compose the projection with $z \mapsto \exp(2\pi \mathrm{i} z/q_j)$. In either case $\chi(c) \neq 1$, so $\chi(x) \neq \chi(y)$. ◀

We next use separation to prescribe values on a finite set.

▶ **Lemma 11** (Finite character interpolation). *Let H be a finitely generated abelian group and let $S \subseteq H$ be finite. The restrictions to S of characters of H span $\mathbb{C}^S$. In particular, if $|S| = m + 1$ and $s_0 \in S$, there are at most $m + 1$ characters and coefficients whose linear combination $g : H \to \mathbb{C}$ satisfies $g(s_0) = 1,\ g(s) = 0 \quad (s \in S \setminus \{s_0\})$.*

Proof. Let $V \subseteq \mathbb{C}^S$ be the span of all character restrictions. It contains the constant function 1 and is closed under pointwise multiplication, since a product of characters is a character. By Lemma 10, it separates distinct points of S.

Fix $x \in S$. For each $y \neq x$, choose $f_{x,y} \in V$ with $f_{x,y}(x) \neq f_{x,y}(y)$, and put

$$h_{x,y}(z) = \frac{f_{x,y}(z) - f_{x,y}(y)}{f_{x,y}(x) - f_{x,y}(y)}.$$

This function belongs to V, equals 1 at x, and equals 0 at y. The product $\prod_{y \neq x} h_{x,y}$ is therefore the indicator function of x on S and belongs to V. The point indicators span $\mathbb{C}^S$, so $V = \mathbb{C}^S$. A basis can be selected from the spanning character restrictions. It has $m + 1$ members, and expressing the indicator of s_0 in this basis gives the required function on H. ◀

3.2 The Product Bound

Character interpolation gives one factor of $m_t + 1$ per coordinate. The following theorem shows how these factors combine.

▶ **Theorem 12** (Ordered coordinate differences). *Let $\Gamma_1, \ldots, \Gamma_k$ be abelian groups. For each t, let $C_t \subseteq \Gamma_t \setminus \{0\}$ be finite with $|C_t| = m_t$. Suppose that an ordered sequence $p_0, \ldots, p_{N-1} \in \Gamma_1 \times \cdots \times \Gamma_k$ satisfies: for every $0 \leq a < b < N$, some coordinate t has $p_b^{(t)} - p_a^{(t)} \in C_t$. Then $N \leq D = \prod_{t=1}^{k} (m_t + 1)$.*

Proof. Let $H_t \leq \Gamma_t$ be the subgroup generated by C_t and $p_0^{(t)}, \ldots, p_{N-1}^{(t)}$. It is finitely generated, even if Γ_t is not. Apply Lemma 11 to $\{0\} \cup C_t$. Choose $g_t(x) = \sum_{j=0}^{m_t} \alpha_{t,j} \chi_{t,j}(x)$ such that $g_t(0) = 1$ and $g_t(c) = 0$ for every $c \in C_t$. If fewer characters are needed, pad the sum with zero coefficients.

For each $a \in \{0, \ldots, N-1\}$, define a function on $H = H_1 \times \cdots \times H_k$ by $G_a(x_1, \ldots, x_k) = \prod_{t=1}^{k} g_t(x_t - p_a^{(t)})$. At its own point, $G_a(p_a) = 1$. If $a < b$, there is a coordinate t with $p_b^{(t)} - p_a^{(t)} \in C_t$, so the corresponding factor vanishes. Hence $G_a(p_b) = 0$ for $a < b$.

The evaluation matrix $Q = (G_a(p_b))_{0 \leq a,b < N}$ is lower triangular with diagonal entries 1. It is nonsingular, so the functions $G_0, \ldots, G_{N-1}$ are linearly independent. Indeed, evaluating a relation $\sum_a \lambda_a G_a = 0$ at all p_b gives $\lambda^{\mathsf{T}} Q = 0$, and thus $\lambda = 0$.

For each t, translation preserves the span of the chosen characters, since $\chi_{t,j}(x_t - p_a^{(t)}) = \chi_{t,j}(x_t)\chi_{t,j}(-p_a^{(t)})$. Expanding each G_a therefore places it in the space spanned by $\Phi_{j_1,\ldots,j_k}(x) = \prod_{t=1}^{k} \chi_{t,j_t}(x_t),\ 0 \leq j_t \leq m_t$ There are D such functions, so this space has dimension at most D. It contains N linearly independent functions, giving $N \leq D$. ◀

The spanning functions need not be independent. Only the upper bound on their number is used. The proof also shows why arbitrary abelian groups cause no difficulty: all characters are constructed on finitely generated subgroups.

▶ **Proposition 13** (Optimality of the sequence bound). *For every choice $m_1, \dots, m_k \geq 1$, the bound in Theorem 12 is attained using finite cyclic groups.*

Proof. Take $\Gamma_t = \mathbb{Z}_{m_t+1}$ and $C_t = \Gamma_t \setminus \{0\}$. Enumerate all D points in the product group, starting with $p_0 = 0$. For $a < b$, the points are distinct, so their difference has a nonzero coordinate. That coordinate belongs to its forbidden set C_t. ◀

For singleton forbidden sets, there is also a direct proof using one separating character per coordinate. We record it to make the original 2^k bound explicit.

▶ **Corollary 14** (Singleton differences). *Let $c_t \in \Gamma_t \setminus \{0\}$ for $t \in [k]$. If, for every $a < b$ in an ordered sequence of N points, some coordinate satisfies $p_b^{(t)} - p_a^{(t)} = c_t$, then $N \leq 2^k$.*

Proof. Let H_t be generated by c_t and the relevant point coordinates. By Lemma 10, choose χ_t with $\chi_t(c_t) \neq 1$. Set $F_a(x) = \prod_{t=1}^{k} \big(\chi_t(x_t) - \chi_t(p_a^{(t)})\chi_t(c_t)\big)$. Then $F_a(p_a) = \prod_t \chi_t(p_a^{(t)})(1 - \chi_t(c_t)) \neq 0$, while $F_a(p_b) = 0$ for $a < b$. The triangular evaluation matrix makes the F_a linearly independent. Expanding the products places them in the span of the 2^k functions $\prod_{t \in S} \chi_t(x_t)$ for $S \subseteq [k]$, where the empty product is 1. Thus $N \leq 2^k$. ◀

3.3 Consecutive Exchanges

To apply Theorem 12, record each exchange block's label change and take prefix sums. An interval change is then the difference of two prefix points.

▶ **Corollary 15** (Exact interval-avoidance threshold). *Let E be finite, let $\psi_t : E \to \Gamma_t$ be labelings, and let $F_t \subseteq \Gamma_t$ be finite. Suppose $B \subseteq E$ satisfies $\psi_t(B) \notin F_t$ for every t. Let $X_1, \dots, X_\ell \subseteq B$ and $Y_1, \dots, Y_\ell \subseteq E \setminus B$ be nonempty sets, pairwise disjoint within each family. Define*

$$B_{i,j} = \left(B \setminus \bigcup_{s=i}^{j} X_s \right) \cup \bigcup_{s=i}^{j} Y_s.$$

If $\ell \geq D = \prod_t (|F_t| + 1)$, some $1 \leq i \leq j \leq \ell$ satisfies $\psi_t(B_{i,j}) \notin F_t$ for every t. The threshold D is best possible for prescribed positive forbidden-set sizes.

Proof. For each s, t, put $\delta_{s,t} = \psi_t(Y_s) - \psi_t(X_s)$ and $C_t = F_t - \psi_t(B)$. Since B is avoiding, $0 \notin C_t$, and $|C_t| = |F_t|$. Define

$$p_0^{(t)} = 0, \qquad p_q^{(t)} = \sum_{s=1}^{q} \delta_{s,t}, \qquad p_q = (p_q^{(1)}, \dots, p_q^{(k)}).$$

Disjointness gives $\psi_t(B_{i,j}) = \psi_t(B) + p_j^{(t)} - p_{i-1}^{(t)}$. If every interval violates a constraint, then for every $0 \leq a < b \leq \ell$, applying value of $\psi_t(B_{i,j})$ to $[a+1, b]$ gives $p_b^{(t)} - p_a^{(t)} \in C_t$ for some t. Theorem 12 yields $\ell + 1 \leq D$, contrary to $\ell \geq D$.

For optimality, use the D points from Proposition 13, with $p_0 = 0$. Take $B = \{x_1, \dots, x_{D-1}\}$ and distinct elements $y_1, \dots, y_{D-1}$ outside B. Set $\psi_t(x_s) = 0$, $\psi_t(y_s) = p_s^{(t)} - p_{s-1}^{(t)}$, and $F_t = \Gamma_t \setminus \{0\}$. The set B is avoiding, but every nonempty interval exchange has a forbidden label in some coordinate. Thus $D - 1$ blocks do not suffice. ◀

▶ Remark 16 (Singleton and one-label cases). For k singleton forbidden sets, Corollary 15 gives the exact threshold 2^k, replacing the factorial threshold in [5, Lemma 5.2]. For two singleton constraints it gives 4, recovering [5, Claim 5.4]. For one labeling with m forbidden values, it gives $m + 1$.

4 Exact Proximity for SIBO Matroids

We now combine the interval bound with the SI-ordering of the differing elements.

▶ **Theorem 17** (SIBO upper bound). *Under the hypotheses of Theorem 1, every basis A has an avoiding basis B with $|A \setminus B| \leq D - 1$.*

Proof. Choose an avoiding basis B minimizing $|A \setminus B|$, and put $\ell = |A \setminus B|$. Suppose that $\ell \geq D$. Apply Lemma 7 to obtain orderings of $B \setminus A$ and $A \setminus B$.

Every interval exchange $B_{i,j}$, as defined in Lemma 7, is a basis, and $|A \setminus B_{i,j}| = \ell - (j - i + 1) < \ell$. The removed elements lie outside A, and the added elements are distinct missing elements of A. Thus, by minimality, no interval basis is avoiding.

Corollary 15, applied to the singleton blocks, gives an avoiding interval exchange. This is a contradiction, so $\ell \leq D - 1$. ◀

The bound cannot be reduced. The next construction extends the uniform-matroid singleton example of [5, Example 5.1] to arbitrary forbidden-set sizes, using only finite cyclic groups.

▶ **Observation 18** (Uniform matroids are SBO). *For two bases of a uniform matroid, pair common elements with themselves and pair the remaining elements arbitrarily. Exchanging any subset of these pairs gives a set of size r, hence a basis. Therefore uniform matroids are SBO and, in particular, SIBO.*

Proof of Theorem 1. The upper bound is Theorem 17. By Example 5.1 [5] and Observation 18 give the matching lower bound within the SIBO class. ◀

▶ **Corollary 19** (Exact universal radii). *For positive $m_1, \ldots, m_k$,*

$$d_{\text{uniform}}(m_1, \ldots, m_k) = d_{\text{SBO}}(m_1, \ldots, m_k) = d_{\text{SIBO}}(m_1, \ldots, m_k) = \prod_{t=1}^{k} (m_t + 1) - 1.$$

In particular, for k singleton constraints these three radii are $2^k - 1$, and for one forbidden set F the radius is $|F|$.

Proof. The class inclusions give the upper bound for all three classes, and the uniform construction gives the same lower bound for all three. ◀

5 Other Matroid Classes

The interval theorem also applies when a matroid provides compatible exchange blocks rather than an SI-ordering. We state the reduction once and then use the known structural bounds.

▶ **Proposition 20** (Weak exchange proximity). *Let* $D = \prod_t(|F_t| + 1)$. *If* M *is weakly* (α, D)*-base orderable, then every basis* A *of a feasible instance has an avoiding basis* B *with* $|A \setminus B| \le \alpha - 1$.

Proof. Choose a closest avoiding basis B. If $|A \setminus B| \ge \alpha$, Definition 8 supplies D compatible blocks $X_i \subseteq B \setminus A$ and $Y_i \subseteq A \setminus B$. Every consecutive interval of these blocks gives a basis. Corollary 15 gives an interval that is also avoiding.

This exchange is strictly closer to A. Indeed, for a nonempty interval I, the removed elements lie outside A, the added elements lie in $A \setminus B$, and

$$\operatorname{dist}\left(A, \left(B \setminus \bigcup_{i\in I} X_i\right) \cup \bigcup_{i\in I} Y_i\right) = |A \setminus B| - \sum_{i\in I} |Y_i| < |A \setminus B|.$$

This contradicts minimality, so $|A \setminus B| < \alpha$. ◀

5.1 Matroids over a Fixed Finite Field

Combining the reduction with the finite-field exchange theorem gives the following bound.

▶ **Corollary 21** (Finite-field proximity). *Fix a prime power* q. *For a feasible instance on a* $\mathrm{GF}(q)$*-representable matroid and any basis* A, *there is an avoiding basis* B *with* $|A \setminus B| \le h(q, D) - 1$. *For* k *singleton constraints the bound is* $h(q, 2^k) - 1$.

Proof. By Theorem 9(i), the matroid is weakly $(h(q, D), D)$-base orderable. Apply Proposition 20 with $\alpha = h(q, D)$. ◀

For singleton constraints, this replaces the argument $\lfloor (e - \frac{1}{2})k! \rfloor$ used in [5, Corollary 5.11] by 2^k. The result uses the same structural function h.

5.2 Sparse Paving Matroids

The elementary exchange bound for sparse paving matroids gives an explicit radius.

▶ **Corollary 22** (Sparse-paving proximity). *For a feasible instance on a sparse paving matroid and any basis* A, *there is an avoiding basis* B *with* $|A \setminus B| \le 2^D - 1$.

For k *singleton constraints the bound is* $2^{2^k} - 1$.

Proof. By Theorem 9(ii), the matroid is weakly $(2^D, D)$-base orderable. Apply Proposition 20 with $\alpha = 2^D$. ◀

We do not claim this sparse-paving bound is optimal. The conjecture that every sparse paving matroid is SIBO [5, Conjecture 6.1] would improve it to $D - 1$.

Algorithm 1 Bounded-Radius Search

Input: An independence oracle for M, a basis A, labelings ψ_t, finite forbidden sets F_t, and a valid proximity radius d.
Output: An avoiding basis, or a certificate that none exists.

```
Set R ← min{d, |A|, |E| − |A|} and compute a_t ← ψ_t(A) for all t;
for i = 0, 1, ..., R do
    foreach X ⊆ A and Y ⊆ E \ A with |X| = |Y| = i do
        Set B ← (A \ X) ∪ Y;
        Query whether B is independent;
        if B is not independent then
            continue;
        end
        Compute b_t ← a_t − ψ_t(X) + ψ_t(Y) for all t;
        if b_t ∉ F_t for every t then
            return B;
        end
    end
end
return "No avoiding basis exists."
```

6 Search and Counting Consequences

A proximity bound reduces the search to a neighborhood of any basis. We give the algorithm for a class with a known radius d. This includes SIBO matroids with $d = D - 1$, as well as the classes in Section 5 with their respective bounds.

The matroid is supplied through an *independence oracle*, which decides whether a queried set is independent. We assume group addition, inversion, and equality are available, and each forbidden set is given as an explicit list. Oracle queries and group operations are counted separately. The algorithm needs no characters or exchange ordering.

Let $n = |E|$, let r be the rank, and let A be a supplied basis. Put

$$R = \min\{d, r, n - r\}, \quad S_R(r, n) = \sum_{i=0}^{R} \binom{r}{i}\binom{n-r}{i}. \tag{1}$$

The rank truncations are valid because the distance between two bases is at most $\min\{r, n-r\}$. The quantity $S_R(r, n)$ counts all r-element sets within distance R of A.

6.1 Bounded-Radius Search

The algorithm enumerates the elements removed from A and the elements added to it. It tests only candidates of size r.

The last return is valid because d is a proximity guarantee for the input class and constraints. The same enumeration with an arbitrary radius only certifies that its neighborhood contains no avoiding basis.

▶ **Proposition 23** (Oracle complexity). *Suppose d is a valid proximity radius. Algorithm 1 finds an avoiding basis or certifies nonexistence using at most $S_R(r, n)$ independence queries. With $K = \sum_t m_t$, it uses $O\big(kr + k(R+1)S_R(r, n)\big)$ group operations and at most $KS_R(r, n)$ equality tests against forbidden values.*

Proof. For each i, there are $\binom{r}{i}\binom{n-r}{i}$ pairs (X, Y), giving exactly $S_R(r, n)$ candidates. Each candidate has r elements, so one independence query determines whether it is a basis. If an avoiding basis exists, the proximity guarantee and rank truncations place one among these candidates. Exhausting the enumeration therefore certifies nonexistence.

Precomputing all $\psi_t(A)$ takes $O(kr)$ group operations. For a candidate at distance i, the identity $\psi_t(B) = \psi_t(A) - \psi_t(X) + \psi_t(Y)$ uses $O(k(i+1))$ group operations over all coordinates. Scanning the explicit forbidden lists uses at most $\sum_t m_t = K$ equality tests. Summing over the candidates gives the bounds. ◀

If membership in each F_t is provided as an oracle, the scans can instead be replaced by at most $kS_R(r, n)$ membership queries. For singleton sets, $K = k$, and the combined group-operation and equality-test bound becomes $O\big(kr + k(R+1)S_R(r, n)\big)$, as in the singleton search. If no starting basis is supplied, the greedy algorithm finds one with at most n additional independence queries.

▶ **Corollary 24** (Search on SIBO matroids). *On a SIBO matroid, Algorithm 1 is correct with $d = D - 1$. Moreover, $S_R(r, n) \le (R+1)(n+1)^{2R}$.*

In the unit-cost oracle model, for fixed forbidden-set sizes the algorithm runs in polynomial time. For singleton forbidden sets, its running time is $(n+1)^{O(2^k)}$.

Proof. The radius follows from Theorem 1. For every $i \le R$, both binomial factors in (1) are at most $(n+1)^i$. Summing gives the displayed estimate. Proposition 23, together with polynomial work to form each candidate, gives the running-time claims. ◀

Thus the singleton search is an XP algorithm parameterized by k: its polynomial exponent depends on k. More generally, the search is XP parameterized by $K = \sum_t m_t$, since $D \le 2^K$. The same applies to the finite-field and sparse-paving radii for fixed q. Neighborhood enumeration alone does not give an FPT algorithm, whose exponent would be independent of the parameter.

6.2 The Number of Avoiding Bases

The same neighborhood size gives a lower bound on the proportion of avoiding bases. This is the multiple-label version of the counting argument in [5, Section 1].

▶ **Corollary 25** (Counting avoiding bases). *Let $\mathcal{B}$ be the bases of a matroid in a class with proximity radius d, and let $\mathcal{B}_{\text{avoid}} \neq \varnothing$ be its avoiding bases. With R and $S_R(r, n)$ as in (1),*

$$|\mathcal{B}_{\text{avoid}}| \ge \frac{|\mathcal{B}|}{S_R(r, n)}.$$

In particular, for SIBO matroids one may take $R = \min\{D - 1, r, n - r\}$.

Proof. For each $A \in \mathcal{B}$, choose an avoiding basis at distance at most R. A fixed avoiding basis can be chosen only by bases in its radius-R neighborhood. Among all r-element subsets of E, that neighborhood contains exactly $S_R(r, n)$ sets, so it contains at most that many bases. Counting the preimages gives $|\mathcal{B}| \le |\mathcal{B}_{\text{avoid}}| S_R(r, n)$. ◀

7 Conclusion

We determined the exact proximity radius for SIBO matroids with multiple abelian-group labelings and finite forbidden sets. For forbidden-set sizes $m_1, \ldots, m_k$, the radius is $\prod_t (m_t +$

1) − 1, and uniform matroids with finite cyclic groups attain it. In particular, the singleton radius is $2^k - 1$, answering the SIBO case of the ICALP 2025 question.

The upper bound follows from character interpolation and a triangular evaluation matrix for prefix points. The same interval bound gives finite-field and sparse-paving consequences. It also gives a direct bounded-radius search and a lower bound on the number of avoiding bases. An important direction is the sparse-paving conjecture from the previous work: if every sparse paving matroid is SIBO, then our exact bound $D - 1$ would also hold for sparse paving matroids. Extending the exact radius beyond SIBO matroids and obtaining faster algorithms remain natural open directions.

AI- Assisted Preparation

The proof ideas and approaches presented in this work were developed through interactive discussions between the authors and ChatGPT (GPT-6.1 Sol), which contributed observations and suggested directions for developing the arguments. The manuscript was composed by the authors, with ChatGPT subsequently assisting in formalizing selected arguments and refining the language and exposition. The authors independently verified all mathematical claims, proofs, and citations, reviewed the final text, and take full responsibility for the content of this work.

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